Mathematics Extended Essay on the topic of Erdős' Probabilistic method

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Contents

How can we use Erdős' Probabilistic method to explore clique sizes in Scale-Free Networks without requiring constructive methods?

1 Introduction

In my endless search for a fitting topic to write my Maths Extended Essay in, I came across the branch of combinatorics. Never having understood the underlying field of study, and knowing that we would cover very little of it in the AI HL syllabus, I was interested in exploring it further. I looked into randomization, graph theory, probabilities, algorithm design and finally landed on Erdős' Probabilistic Method. Fascinated by the concept of non-constructive proofs, my understanding followed closely by my curiosity increased rapidly which led me to what I quickly realised was the frontier of mathematics. Only understanding very little but eager to learn I continued researching. This extended essay is a summary of these findings and an attempt at combining two inherently close concepts that I could not find any information on in my research: how clique sizes relate to the power-law distribution of scale free networks.

This essay will be structured into three main parts. First, I will explain the knowledge I acquired through my research to a level a Maths AI HL student can understand. I will look at the concept of binomial coefficients, the generation of random graphs and specifically scale free networks, and present the basics of the probabilistic method. Second, using these tools, I will derive a relationship between the exponent γ in distribution of node degrees and the clique sizes found in a predefined graph. Finally, I will empirically confirm my derivation through the generation of scale free networks to generate statistically significant results.

To briefly help the reader to understand this essay better, below are a few definitions and pointers on notation which might be useful throughout the next few pages:

Object The concept of an object is a highly philosophical matter, yet for the purpose of this essay, the reader can assume that it is simply an abstract *thing*. It can be anything from a number, a node of a graph or a name on a list.

Set A set is a collection of such *things*, i.e. objects.

Element An element is a distinct object in a set.

$\mathbf{s} \in \mathbf{S}$ This notation ($\in$) will come up frequently and means "s is an element of S".

Binomial Coefficient The notation of a binomial coefficients will be discussed in the next section.

2 Combinations and Permutations

To understand the rest of this essay, it is important to clarify the concept of what are colloquially known as *combinations*. Imagine a simple cake in which all ingredients are mixed independent of the order. In mathematical language, that is called a combination of objects. If, on the other hand, the order matters, like for a phone password, that is called a permutation. There is a further distinction between a combination and permutation where repeats are allowed or not. A phone password, for example, can have repeating digits, arranging the winners of a race, on the other hand, cannot.

2.1 Permutations with Repetitions

Calculating possible arrangements of objects depends on the requirements. It is simplest to calculate all possible permutations when repetitions are allowed. If we have a 4-digit phone password where every digit can be a digit from 0-9 with repetitions there are:

$$10 \times 10 \times 10 \times 10 = 10^4 = 10000$$

possible permutations since every new digit we can choose from another set of 10 numbers.

Generalising, if we have n different types in a specific order of r items with repetitions allowed, we can assume there are:

$$n^r \tag{1}$$

permutations.

2.2 Permutations without Repetitions

If, however, we cannot have repeating values, because we are working with all the possible results of a race, we have to decrease the pool of objects to pick from with every step. When repetitions were allowed, we could multiply the same number over and over. This time, however, we need to calculate the possibilities from a decreasing number of choices. Taking 7 runners in a race, there are 7 options to be first, but then only 6 to come second, then 5 and so on. There are therefore

$$7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 =$$

possibilities. Mathematically, this can be denoted as 7!. If want to choose the first three winners only we would multiply $7 \times 6 \times 5$. Since our notation goes until 1 however, we can cancel out the latter part:

$$\frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1} = \frac{7!}{4}$$

Generalising, if we chose r items from n different types where repetitions are not allowed we get:

$$\frac{n!}{(n-r)!} \tag{2}$$

If we apply this idea to combinations where there are no repetition, in other words we do not care about the order but cannot have duplicates, we need to remove all duplicates. Taking our race example, if we only care about the all the possible lists we can generate, independent of their order, we have to remove all the ways in which 3 objects (the value from before) can be arranged. As a reminder, that is 3!. Now applying that to the formula:

$$\frac{\text{number of possible combinations}}{\text{cancelling out} \times \text{eliminating duplicates}} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{(4 \times 3 \times 2 \times 1) \times (3 \times 2 \times 1)} = \frac{7!}{4! \times 3!}$$

Conventional notation switches the latter around meaning we generalise to:

$$\frac{n!}{r!(n-r)!}$$

This formula is so common that it is shortened to:

$$\frac{n!}{r!(n-r)!} = \binom{n}{r} \tag{3}$$

The notation means "n chooses r" and we are taking r items from a pool of n different types where the order does not matter, but there cannot be repeats. This formula is called the binomial coefficient and will be used frequently throughout this essay.

3 Scale Free Networks and Graph Theory

- Describing the concept of graphs - Defining terms - Applying the idea to scale free networks

One predominant branch of Mathematics which is often ignored is called combinatorics. It focuses on the idea of counting, which is applied in order to find a result but can also define a final answer. I will talk more about that concept using an example below. Part of combinatorics is the concept of graphs which is related to graph theory which explores the relationship between objects. These should not be confused with a graph on a Cartesian plane. These graphs allow the mathematical analysis of networks and connections, like streets, mobile networks, or social media networks. From now on the use of the word graph will refer to the mathematical graph which makes part of discrete mathematics, i.e. a network of nodes connected by edges.

3.1 Definitions

For clarity, I will define the following terms, as they are different around the mathematical world:

Node: A node is the same as a vertex and is the undivisible point at which one or more edges meet.

Edge: An edge is the connection between nodes.

Degree: The degree of a node is the number of connections, i.e. edges, it has to another node. In general, this essay will denote it as k to simplify understanding.

Complete Graph A graph in which every pair of distinct nodes is connected by an edge. It will be denoted as K_n where n is the number of nodes in the graph.

Subgraph A subgraph by which is most commonly meant induced subgraph is a subset of nodes of a graph and all the edges connecting pairs of the subset. This creates a new graph "inside" another graph.

Clique A clique is a tighter definition of a subgraph as it requires the subgraph to be complete.

I am only making use of simple undirected graphs, meaning that there can only be a single edge between every node and that the edges do not represent a direction in the connection between nodes (as would be required when representing one-way streets for example).

3.2 Scale Free Networks

This extended essay will investigate scale free networks which are representative of many real-world networks because of their "Hubs". A scale free network gives a power a relation between the degree of a node and the probability of its occurrence. In other words,

$$P_{(deg)}(k) \propto k^{-\gamma} \tag{4}$$

where γ is some exponent and $P_{(deg)}(k)$ the probability of a node with degree k.

To generate scale free networks, different methods exist. The greater category of the most commonly used model is called *configuration model* because it allows the degree of every node to be pre-defined. Since scale free networks are defined by their degree separation that is a requirement. When it is of interest to fix the degree sequence exactly it is possible to make use of a microcanonical model which defines the exact degree of every node in the graph and randomly draws edges according to these requirements. On the other hand, a canonical model works with the probability distributions of the edges in a network. The one this extended essay focuses on is the Chung-Lu (canonical) Configuration Model because the probabilistic generation is also easier to analyse using the probabilistic method.

In short, the model defines the probability of an edge between two nodes h and j to be:

$$p_{hj} = \frac{k_h k_j}{\sum_i k_i} = \frac{k_h k_j}{2m} \tag{5}$$

where p_{hj} is the probability, k_h and k_j are their respective degrees (remember: the number of nodes they are connected to). We divide by the sum of all degrees to make a probability that sits between 0 and 1.

Equation (5) comes from the bounds set by the definitions of algebraic mathematics and graph theory. Chung & Lu explicitly define the graph G($\mathbf{w}$) where $\mathbf{w}$ is the expected degree sequence. The expected degree sequence of a graph is the decreasing sequence of its nodes' degrees. In the graph below that would be:

They further define that $G \in G(\mathbf{w})$ to show that we are not talking about a subgraph. To make the model work, it is further stated that all probabilities are independent and that for the greatest i, $k_i^2 < \sum_i k_i$. In other words, if we take two nodes with the greatest degrees, their product cannot be greater than the sum of all degrees.

To apply this graph to scale free networks, we have to remember the relation from equation (4) and weigh the probability accordingly.

4 The Probabilistic Method

The exploration of random graphs which we discussed above is possible through many different methods due to the graphs' statistical nature. For every graph generated, it is possible to compute different values such as the diameter, which is the maximum operation between two nodes or the girth, which is the shortest cycle in the graph (where cycle means starting from one node travelling along the graph and returning to the original node without passing through the same node twice). It is also possible to compute certain clique sizes. Although straightforward to conceptualise, the greater the number of nodes the more complicated these calculations become.

The method I am going to apply is called the *Probabilistic Method*. Pioneered by Paul Erdős, it is used to prove the existence of mathematical objects non constructively. By non constructively I mean that by the end of the proof we only know that an object has a non-zero probability of existing, yet we do not know where it is.

To help the reader understand, I am going to lay out a proof using the probabilistic method first published by Erdős in 1947. This proof requires understanding the concept of edge colouring and makes use of Ramsey numbers.

4.1 Edge Colouring

Start with edge colouring. Imagine there is a neighbourhood party which you have been invited to. You know some of you neighbours and others you have barely seen and yet you interact with all of them throughout the evening. If we want to demonstrate that using a graph, every node would be a neighbour, and every edge an interaction. In this example graph colouring allows us to define the type of relation between two nodes connected by an edge, i.e. a known or an unknown neighbour. When optimising for edge colouring to avoid two edges of the same colour to be connected to a node, such a representation can be used in classroom bookings for study or frequency assignments of transceivers.

4.2 Ramsey Numbers

Another application of edge colouring are Ramsey numbers $R(k, l)$. They are defined by the smallest number of nodes, n, in a complete graph K_n coloured red and green required to guarantee a monochromatic red clique of node-size K_k or monochromatic green clique node size K_l. Taking this apart, we are asking how many interconnected nodes we need to randomly generate, where every edge is coloured either red or green with any probability, to find a pattern in seeming chaos. In our case this pattern is a complete subgraph, i.e. clique, which has a single colour either red or green connecting k or l nodes respectively. The reader should note that the colours are arbitrarily chosen.

The most common example is $R(3, 3) = 6$. Below is a randomly generated graph with 6 nodes and random colouring, with the green triangle (clique of size 3 makes a triangle) clearly visible between 1, 3 and 5.

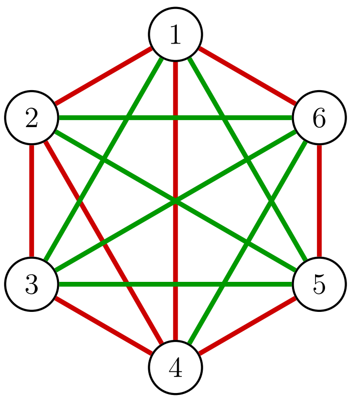

Figure 1:

On the other hand, below is a graph with 5 nodes where there are no monochromatic cliques, showing that $R(3, 3) > 5$:

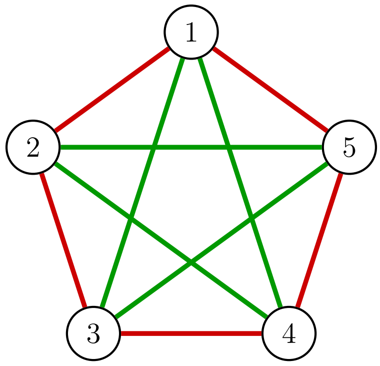

Figure 2:

4.3 Finding Lower Bounds of Ramsey Numbers

To return to the probabilistic method and with the concept of Ramsey numbers explained here is the proof published by Erdős in 1947:

Theorem 1 (Erdős 1947)**.** *If* $\binom{n}{k} \times 2^{1-\binom{k}{2}} < 1$ *then* $R(k,k) > n$

Proof. Let's start by colouring the edges of a complete graph on K_n nodes using two colours uniformly at random. The edges are coloured of equal probability:

$$P(\text{an edge is red}) = P(\text{an edge is green}) = \frac{1}{2}$$

There are k (from our diagonal Ramsey number $R(k,k)$) nodes from n total nodes we want to pick from meaning we can apply the formula from section 2:

$$\frac{n!}{r!(n-r)!} = \binom{n}{r} = \binom{n}{k}$$

Next, let's define an event, A_S in which a clique S of size k is monochromatic. In other words, a subgraph which has k nodes of our initial graph K_n of a single colour. The probability of that occurring is:

$$P(A_S) = 2 \times (\frac{1}{2})^{\binom{k}{2}}$$

Where the leading two is there because we do not care about which of the two colours the clique has. The $\frac{1}{2}$ stems from the initial probability of the colouring of the edges. And the $\binom{k}{2}$ defines all possible edges in the clique, i.e. all pairs of nodes. We can simplify this to:

$$P(A_S) = 2 \times (\frac{1}{2})^{\binom{k}{2}} = 2 \times 2^{-\binom{k}{2}} = 2^{1-\binom{k}{2}}$$

Since there are $\binom{n}{k}$ possible arrangements of the clique size k, we can then multiply that to find the probability of any clique of size k of any colour to be monochromatic:

$$P(\text{there exists a monochromatic clique } K_k) = \binom{n}{k} \times 2^{1-\binom{k}{2}} \tag{6}$$

By the assumption of the theorem, however, we know that:

$$\binom{n}{k} \times 2^{1-\binom{k}{2}} < 1$$

And therefore there is a positive probability that there exists such a graph for a complete graph K_n:

$$P(\text{there does not exist a monochromatic clique } K_k) > 0$$

If that is the case, we have not yet achieved the requirements of a Ramsey number which is the absence of the possibility that *there does not exist a monochromatic clique* which is why:

$$R(k, k) > n$$

For n which we chose at the beginning of the proof.

□

5 Predicting clique sizes in Scale Free Networks

With both the concepts of scale free networks and the probabilistic method explained, we can now move on to the core of this essay.

6 Building Scale Free Networks

7 Comparing Results

- Comparing results and making sure theory aligns with empirical data

8 Conclusion

- Conclude on the uses of my results